



# Moments and Equilibrium of Rigid Bodies

## Worksheet 1

Mechanics Chapter 5: Moments and Equilibrium of Rigid Bodies. Answer all questions. Show your working clearly. Total: 62 marks. Time: 80 minutes.

Name \_\_\_\_\_

Class \_\_\_\_\_

Date \_\_\_\_\_

### INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 5. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take  $g = 9.8 \text{ ms}^{-2}$ . Moments are measured in N m; say which point you take moments about. Give answers to 3 significant figures.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

| Section                       | Marks available | Marks scored |
|-------------------------------|-----------------|--------------|
| A Moments of a force          | 16              |              |
| B Equilibrium of a rigid body | 21              |              |
| C Sliding and toppling        | 20              |              |
| <b>Roar-level challenge</b>   | 5               |              |
| <b>Total</b>                  | <b>62</b>       |              |

### Section A Moments of a force

16 marks

1. A force of 12 N acts at right angles to a spanner, 0.5 m from the nut. Find the moment of the force about the nut. [2]

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2. A light rod  $AB$  has length 4 m. Vertical forces of 5 N and 8 N act downwards on the rod at distances 1 m and 3 m from  $A$ . [2]  
Find the total moment of these forces about  $A$ , stating its sense.

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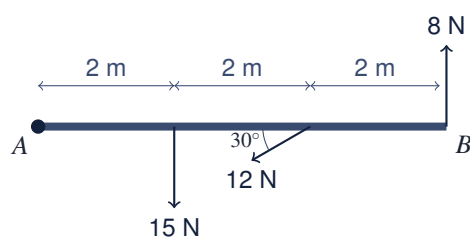
3. A force of 20 N acts at a point 2 m from a pivot  $O$  on a rod. The force makes an angle of  $30^\circ$  with the rod. [3]  
Find the moment of the force about  $O$ .

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4. A light rod  $AB$  of length 6 m is freely pivoted at  $A$  and lies horizontally. The forces shown act on the rod. [3]



Find the resultant moment about  $A$ , stating whether it is clockwise or anticlockwise.

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5. A light rod  $AB$  of length 3 m lies horizontally. A downward force of 20 N acts at  $C$ , where  $AC = 1$  m; a downward force of  $F$  N acts at  $B$ ; and an upward force of 30 N acts at  $D$ , where  $AD = 2$  m.

(a) Given that the resultant moment about  $A$  is zero, find  $F$ .

[3]

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(b) Find the resultant force on the rod.

[2]

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(c) Explain why the rod is not in equilibrium, even though the moment about  $A$  is zero.

[1]

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*(Total for Question 5 is 6 marks)*

**Section B   Equilibrium of a rigid body**

21 marks

6. A uniform rod of length 2 m and mass 4 kg rests horizontally on two supports, one at each end. Find the reaction at each support. [2]

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7. A uniform plank  $AB$  of length 4 m and mass 20 kg rests horizontally on two supports, one at  $A$  and one at  $C$ , where  $CB = 1$  m. Find the reactions at  $A$  and  $C$ . [3]

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8. A uniform see-saw of length 3 m is pivoted at its midpoint. A child of mass 30 kg sits 1.2 m from the pivot. Find where a child of mass 24 kg must sit for the see-saw to balance. [3]

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9. A non-uniform rod  $AB$  of length 5 m and mass 10 kg rests horizontally on supports at  $A$  and  $B$ . The reaction at  $A$  is 40 N. Find the distance of the centre of mass of the rod from  $A$ . [4]

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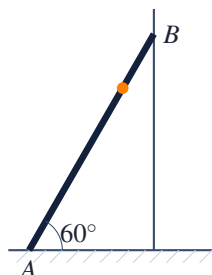
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10. A uniform ladder  $AB$  of length 6 m and mass 20 kg rests with  $A$  on rough horizontal ground and  $B$  against a smooth vertical wall. The ladder makes an angle of  $60^\circ$  with the ground. A man of mass 80 kg stands on the ladder at a point 4.5 m from  $A$ . The ladder is in equilibrium.



- (a) Find the normal reaction at  $A$ . [1]

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- (b) By taking moments about  $A$ , find the reaction at  $B$ . [4]

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- (c) Find the least possible value of the coefficient of friction between the ladder and the ground. [3]

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- (d) State one modelling assumption you have made about the man. [1]

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(Total for Question 10 is 9 marks)



## Section C Sliding and toppling

20 marks

11. A box of weight 100 N rests on rough horizontal ground, with coefficient of friction 0.5. A horizontal force of 40 N is applied. Determine whether the box moves. [2]

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12. A uniform block has a square base of side 0.6 m, height 1.2 m and weight 150 N. It stands on rough horizontal ground. A horizontal force  $P$  is applied at the top edge, perpendicular to a face. Assuming the block does not slide, find the value of  $P$  at which it is about to topple. [3]

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13. For the block in Question 12, the coefficient of friction is 0.3. As  $P$  is gradually increased from zero, determine whether the block slides or topples first. [3]

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14. A uniform rod  $AB$  of length 4 m and mass 10 kg rests horizontally on supports at  $C$  and  $D$ , where  $AC = 0.5$  m and  $DB = 1$  m. A particle of mass 15 kg is placed on the rod between  $D$  and  $B$ . Find the greatest distance from  $D$  at which the particle can be placed without the rod tilting. [4]

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15. A uniform plank  $AB$  of length 6 m and mass 30 kg rests horizontally on two supports at  $C$  and  $D$ , where  $AC = 1$  m and  $DB = 2$  m. A man of mass 75 kg walks along the plank from  $A$  towards  $B$ .

(a) Find the reactions at  $C$  and  $D$  when the man stands at the midpoint of the plank. [3]

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(b) Find the closest distance to  $B$  at which the man can stand before the plank starts to tilt. [3]

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(c) A child sits at  $A$  so that the man can just reach  $B$  without the plank tilting. Find the mass of the child. [2]

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(Total for Question 15 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. A uniform ladder rests with its foot on rough horizontal ground (coefficient of friction  $\mu$ ) and its top against a smooth vertical wall. The ladder makes an angle  $\theta$  with the ground and is in limiting equilibrium. [5]

Show that  $\tan \theta = \frac{1}{2\mu}$ , and find  $\theta$  when  $\mu = 0.4$ .

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END OF WORKSHEET

Total: 62 marks